The Effects of Recreational Marijuana Legalization on College Retention Rates and Graduation Rates

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Abstract

The paper studies the impacts of the legalization of recreational marijuana use on college retention rates and graduation rates in the U.S. by estimating a differences-in-differences model. Data used in the paper come from over $5,095$ U.S. colleges who have reported retention rates and graduation rates from year 2008 to year 2020 in the Integrated Postsecondary Education Data System (IPEDS). I find that state recreational marijuana legalization decreased the college retention rates in the fourth year after the legalization by 1.5 percentage points. Normal-time graduation rates significantly decreased in the first, fourth and sixth year after the legalization by 2.2 percentage points, 3.2 percentage points and 3.9 percentage points respectively.

1 Introduction

Recreational marijuana has been legalized for adults 21 years of age or older in several states beginning with Colorado and Washington in 2012. As of July 1, 2022, adult recreational use of marijuana has been legalized in 19 states, 2 territories(Guam and the Northern Mariana Islands), and the District of Columbia. While there is growing acceptance of marijuana use, the effects of marijuana to the society are still uncertain.

College-aged students is a population generally thought to be predisposed towards risky behaviors, and marijuana is among the drugs most commonly used by young adults (ages 19-30). According to the 2021 National Survey Results on Drug Use of Monitoring the Future (MTF), in 2020, over four-in-ten young adults (42%) used marijuana at least once in the past 12 months, over one-in-four (27%) used it at least once in the past 30 days, and nearly one-in-ten (9.8%) was a daily or near-daily marijuana user in the past 30 days (Johnston et al., 2022). A bunch of existing studies have shown the recreational marijuana legalization (RML) increased the marijuana consumption among college students. Miller et al. 2017 paper is the earliest literature assessing the impact of recreational legalization on college students' marijuana use based on the legalization in Washington. Using survey data from the National College Health Assessment, they show that students at Washington State University experienced a significant increase in marijuana use after legalization (Miller et al., 2017). Jones et al. 2018 paper provides evidence based on Colorado, the first sate to legalize recreational marijuana use, and shows that the frequency of marijuana use after RML in Colorado college students is much higher than the national average (Jones et al., 2018). Later, evidence based on state of Oregon also shows that students' rates of marijuana use have increased following recreational marijuana legalization (Koval et al., 2019). Most recent paper from Barker and Moreno uses a longitudinal study to evaluate the effects of legalizing recreational use on the attitudes, intentions, and marijuana use behaviors of college students in two different legalization contexts, Washington State and Wisconsin. Their results show that, among prior users, the proportion using in the last 28 days rose faster in Washington after legalization that it did in Wisconsin (Barker and Moreno, 2021).

However, marijuana use is associated with increased cognitive, mental health, and substance use disorders, including difficulties with verbal performance, attention, working memory, and inhibition (Ammerman et al., 2015). And many of these health concerns fall disproportionately on young users. This paper studies the effects of recreational marijuana legalization on college students' education outcomes, retention rates and graduation rates.

Besides the impacts on marijuana consumption, there are studies on whether marijuana serve as a substitute or complements for other substances. Economic theory has focused on the association between alcohol and marijuana use with some authors suggesting that alcohol and marijuana use are complements, while others suggest they are substitutes. For example, Pacula (1998) reported that an increased federal tax on beer had " a negative and significant effect on the demand for both alcohol and marijuana, implying that alcohol and marijuana are economic complements" (Pacula, 1998).In contrast, other groups have shown that restricting the availability of alcohol by increasing the minimum drinking age is associated with increased marijuana use and a decrease in marijuana use and increase in alcohol use around the age of 21 years, when alcohol use becomes legal, consistent with the substitute model (Crost and Guerrero, 2012). Marijuana and tobacco use are also highly comorbid and some recent work suggests that changes in marijuana use may impact tobacco use, though the directionality of the relationship also remains unclear. For example, Allsop et al. (2014) demonstrated that during voluntary abstinence from marijuana use, tobacco use increased by 14 cigarettes per week(Allsop et al., 2014). In contrast Peters & Hughes (2010) studying daily marijuana users during a 13 day period of abstinence, did not demonstrate increases in cigarette or non-marijuana illicit drug use(Peters and Hughes, 2010). Finally, it has recently been suggested that medical marijuana use may be associated with reductions in opioid use (Nunberg et al., 2011).

Based on previous studies, the effects on recreational marijuana legalization affect people both in positively ways and negative ways, and the results are mixed. Solely based on these first-stage studies, it is hard to determine the impacts of the legalization to the society. While there are also studies on the intent-to-treat effects of RML on crime (Wu et al., 2021), employment (Chakraborty et al., 2021) and traffic safety (Farmer et al., 2022), there a dearth of study link the RML to education. One possible reason is that the limitation of longitude individual-level data on education outcomes.

In this paper, I study the effects of recreational marijuana legalization on two important education outcomes, college retention rates and graduation rates. I use institution-level data of retention rates and graduation rates of $5,095$ U.S. postsecondary institutions from the 2008 to 2020 Integrated Postsecondary Education Data System (IPEDS) who have reported the retention rates and graduation rates during the period. The sample covers both 4-year institutions and two-year and less-than-2-year institutions, as well as public institutions, private not-for-profit institutions, and private-for-profit institutions. To identify the causal effects, I use an event-study model which can both test the common-trend assumption and estimate heterogeneous treatment effects.

Results from my analysis show the college retention rates dropped significantly by 1.5 percentage points in the fourth year after state legalizing the recreation use of marijuana, and the effects is mainly driven by 4-year institutions. In terms of the graduation rates, while there aren't any significant effects on graduation rates within 150% normal time and 200% normal time, it decreased the normal-time graduation rates significantly in the first, fourth and sixth year after the state legalized recreational marijuana use by 2.2 percentage points, 3.2 percentage points and 3.9 percentage points respectively. Thus, I concluded that RML potentially extended college students' graduation time.

2 Recreational Marijuana Legalization Overview

As of July 1, 2022, adult recreational use of marijuana has been legalized in 19 states, 2 territories(Guam and the Northern Mariana Islands), and the District of Columbia. Table

1 lists the time and process of recreational marijuana legalization (RML) in the U.S..

Colorado and Washington approved adult-use recreational marijuana measures in 2012. Alaska, Oregon and District of Columbia followed suit in fall of 2014. In 2015, Ohio voters defeated a ballot measure that addressed commercial production and sale of recreational marijuana. On Nov 8, 2016, voters in four states, California, Maine, Massachusetts and Nevada, approved adult-use recreational marijuana, while voters in Arizona disapproved. In 2018, Michigan voters approved "Proposal 1" by a margin of 56 percent to 44 percent to legalize, regulate, and tax marijuana in the state. I n 2018, Vermont became the first state to legalize marijuana for adult use through the legislative process (rather than a ballot initiative.) Vermont's law went into effect July 1, 2018. In May 2019, the Illinois General Assembly passed the Cannabis Regulation and Tax Act, House Bill 1438 and the Governor signed the legislation in June. Ballot measures legalizing recreational marijuana were approved by voters at the November 2020 election in Arizona, Montana and New Jersey. In the 2020 elections, South Dakota voters passed Constitutional Amendment A (54%). Quickly after the amendment passed, it was challenged in court. On Feb. 8, 2021 Circuit Judge Christina Klinger ruled that the measure was unconstitutional. New York legalized cannabis with the governor's signature of AB 1248, which was passed during New York's 2021 session. Additionally, on March 30, 2021, New Mexico's legislature introduced its Cannabis Regulation Act within their first special session. The bill, HB 2a, has been signed by the Governor. On April 7, 2021, Virginia's legislature accepted the governor's recommended amendments for a significantly speedier implementation window for HB 2312 which would legalize recreational cannabis use in Virginia and establish a regulated commercial market. In the same year, Connecticut legalized the recreational use of cannabis with the passage of SB 1201. Some aspects of the bill do not take effect immediately, but those 21 years old and higher will be able to recreationally use cannabis starting July 1, 2021. Moat recent, Rhode Island Gov. Dan McKee signed into law the Rhode Island Cannabis Act on May 25 2022, making Rhode Island the 19th state to legalize marijuana for recreational use.

Table 1. Timeline and Process of Recreational Marijuana Legalization

	State	Year	Process Used	How Passed	Possession Limits
1	Colorado	2012	Initiative	Amendment 64 (55%)	1 oz usable; 6 plants (no more than 3 mature); 1 oz hash/concentrates
2	Washington	2012	Initiative	Initiative 502 (56%)	1 oz usable; 16 oz solid marijuana-infused, 72 oz liquid infused, and 7 g of concentrates
3	Alaska	2014	Initiative	Ballot Measure 2 (53%)	1 oz usable; 6 plants (no more than 3 mature)
4	Oregon	2014	Initiative	Measure 91 (56%)	1 oz usable in public; 8 oz homegrown usable at home; 4 plants; 16 oz solid marijuana-infused, 72 oz liquid infused, and 1 oz extract at home of hash/concentrates
5	Washington, D.C.	2014	Initiative	Initiative 71 (65%)	2 oz usable; 6 plants (no more than 3 mature)
6	California	2016	Initiative	Proposition 64 (57%)	1 oz usable; 6 plants; 8 g hash/concentrates
7	Maine	2016	Initiative	Question 1 (50%)	2.5 oz usable; up to 15 plants (no more than 3 mature); 5 g hash/concentrates
8	Massachusetts	2016	Initiative	Question 4 (54%)	1 oz usable; 6 plants; 5 g concentrates
9	Nevada	2016	Initiative	Question 2 (54%)	1 oz usable; 6 plants; 3.5 g hash/concentrates
10	Michigan	2018	Initiative	Proposal 1 (56%)	2.5 oz usable; 12 plants; 15 g concentrates
11	Vermont	2018	Legislation	Legislative Bill H.511	1 oz usable; 6 plants (no more than 2 mature); 5 g hash
12	Illinois	2019	Legislation	House Bill 1438	1 oz usable; 5 g hash/concentrates
13	Arizona	2020	Initiative	207) (59.95%)	1 oz usable; 6 plants
14	Montana	2020	Initiative	Initiative I-190 (56.89%) and CI-118 (57.82%)	1 oz usable; 4 mature plants; 8 g hash/concentrates
15	New Jersey	2020	Referral	Legalization Amendment (66.88%)	1 oz usable
16	New York	2021	Legislation	Senate Bill S854A	3 oz usable; 12 plants; 24 g concentrates
17	Virginia	2021	Legislation	Senate Bill 1406 House Bill 2312	1 oz usable; 4 plants
18	New Mexico	2021	Legislation	House Bill 2	2 oz usable; 6 plants (no more than 12 per household); 16 g concentrated marijuana; 800 milligrams of edible cannabis
19	Connecticut	2021	Legislation	SB 1201	1.5 oz usable; 6 plants (no more than 3 mature beginning July 2023); up to 7.5 g concentrates (up to 750 mg of THC); up to 25 g concentrates (up to 2,500 mg of THC) in a locked container
20	Rhode Island	2022	Legislation	Rhode Island Cannabis Act	1 oz usable in public; 10 oz usable at home; 3 mature plants and 3 immature plants; 5 g concentrate

3 Data

The empirical analysis in this paper relies on institution-level data from the Integrated Postsecondary Education Data System, or IPEDS. IPEDS is a system of interrelated surveys conducted annually by the National Center for Education Statistics, or NCES, to describe postsecondary education in the United States. It gathers information from every college, university, and technical and vocational institution that participates in the federal student financial aid programs. The Higher Education Act of 1965, as amended, requires that institutions that participate in or apply for these federal student aid programs report to IPEDS. The data collected by IPEDS cover basic characteristics of institutions, enrollments, completions and completers, graduation rates and additional outcome measures, faculty and staff, finances institutional prices, student financial aid, admissions, and academic libraries, which provide basic information on postsecondary institutions for organizations and individuals.

The IPEDS universe includes all institutions that are eligible for federal student aid. This includes institutions of all levels and sectors, as well as both degree granting and non-degree-granting institutions. The sample used in this paper includes $5,095$ U.S. post-secondary institutions, covering both 4-year institutions and two-year and less-than-2-year institutions, as well as public institutions, private not-for-profit institutions, and private-for-profit institutions. The detailed summary statistics for the classifications of the sample is showed in Table 2.

Table 2. Summary Statistics of Institution Classifications

	All Institutions	4-year	2-year	Less than 2-year
Public	1,743	602	919	222
Private Not-for-profit	1,378	1,201	129	48
Private For-profit	1,974	179	531	1,264
Observations	5,095	1,982	1,579	1,534

Main outcome variables of this paper are retention rates and graduation rates. Following the definitions of the IPEDS data, the full-time retention rate in this paper is defined as the percent of the fall full-time cohort from the prior year minus exclusions from the fall full-time cohort that re-enrolled at the institution as either full- or part-time in the current year. Full-time fall cohort includes full-time first-time degree/certificate seeking students enrolled at the institution in the fall of the prior year (including those enrolled for the first time the preceding summer term and those whose intent was not known upon entry to the institution). Exclusions are the number of students from the prior year cohort, who left the institution for any of the following reasons: Died or were totally and permanently disabled; Serve in the armed forces (including those called to active duty); Serve with a foreign aid service of the Federal Government (e.g., Peace Corps); or Serve on official church missions.

The graduation rate is the percentage of students from the adjusted bachelor's degree-seeking cohort, who completed a bachelor's degree within the given period of time. Adjusted bachelor's degree-seeking cohort is calculated as the revised cohort minus exclusions.[1]. In this paper, I conducted separated regressions for graduation rates within 100% of normal time, graduation rates within 150% of normal time, and graduation rates within 200% of normal time. Normal time is 4 years for a bachelor's degree in a 4-year institutions, 2 years for an associate's degree in a standard term-based institution, and the various scheduled times for certificate programs.

Table 3. Summary Statistics of Treated Group and Control Group

	Sample Mean	Treated Mean	Control Mean
Outcome Variables:			
Graduation Rate within 100% of Normal Time	0.37	0.39	0.37
Graduation rate within 150% of Normal Time	0.53	0.55	0.52
Graduation rate within 200% of Normal Time	0.56	0.58	0.55
Full-time Retention Rate	0.71	0.73	0.70
Others:			
Student-to-faculty Ratio	15.76	16.48	15.31
Percent of Full-time First-time Undergraduates Receiving any Financial Aid	85.60	82.15	87.71
12-month Unduplicated Headcount	5559.39	6533.07	4963.74
Historically Black College or University	0.02	0.00	0.03
Male-to-female Ratio	2.07	2.35	1.90
Total Price for In-state Students Living On Campus	32925.86	37131.29	30454.81
Percent Admitted	68.72	67.09	69.78
Adimission Yield	44.87	43.79	45.56
Full-time First-time Degree/certificate-seeking Undergraduate	783.50	850.67	742.18

Data used to generated the table are from IPEDS.
The means of graduation rates, retention rates, student-to-faculty ratio, percent of full-time first-time undergraduates receiving any financial aid, 12-month unduplicated headcount, male-to-female ratio, total price for in-state students living on campus, percent admitted, admission yield, and full-time first-time degree/certificate-seeking undergraduate showed in the table are 13-year averages from year 2008 to 2020. Historically Black College or University is a binary variable which equals to 1 if yes, otherwise 0.

[1]See IPEDS website for more detailed definitions.

The paper used institution-level data of IPEDS from year 2018 to year 2020. Institutions in the sample can be divided into treated group and control group according to the recreational marijuana legalization status of the states they located in. Table 3 shows a summary statistics of the sample by group. It includes both the outcome variables and other main characteristics of the institutions, which implies a balance between the treated group and control group.

4 Identification Strategy: Differences-in-Differences

The research design exploits the effects of the state recreational marijuana use legalization status on college retention rates and graduation rates in a difference-in-differences model that compares institution-level outcomes before and after legalizing of the recreational marijuana use between institutions in legalized states and other states, or in the treated group and control group. The primary identifying assumption of this design is the parallel trend assumption that, in the absence of recreational marijuana legalization, outcomes of institutions in the treatment group would have paralleled outcomes of the institutions in the control group. Equation (1) describes an event study specification (Jacobson et al., 1993).

$$Y_{ist} = \alpha + \boldsymbol{x'_{ist}}\boldsymbol{\beta} + \gamma_s + \delta_t + L_s * \left[\sum_{j=-m}^{-2} \omega_j \mathbb{1}\left\{t - t_s^* = j\right\} + \sum_{j=0}^{n} \theta_j \mathbb{1}\left\{t - t_s^* = j\right\} \right] + \varepsilon_{ist}, \quad (1)$$

Y_{ist} is the outcome variable, retention rates or graduation rates, of institution i in state s of year t. $\boldsymbol{x_{ist}}$ is a vector of institution characteristics, including the indicator variable of whether the institution is a historical black college, category variables of the classification of whether an institution is operated by publicly elected or appointed officials or by privately elected or appointed officials and derives its major source of funds from private sources (public, private not-for-profit, or private for-profit), the level of the institution (4-year, 2-year, or less than 2-year), and the highest degree offered, student-to-faculty ratio, percent of full-time first-time undergraduates receiving any financial aid, institution size (measured by 12-month unduplicated headcount), and percent indicator of undergraduates formally registered as students with disabilities. γ_s captures state fixed effects, and δ_ts are year fixed effects which control for any outcome effects that are common to all institutions in calender year t. Treatment/control groups are defined by the binary variable, L_s, which equals to 1 for institutions in states have legalized recreational marijuana use (treatment group), otherwise (control group) 0.Pre/post treatment is defined by dummy variables that measure the time relative to the state recreational marijuana legalization, $\mathbb{1}\{t - t_s^* = j\}$ (event year).

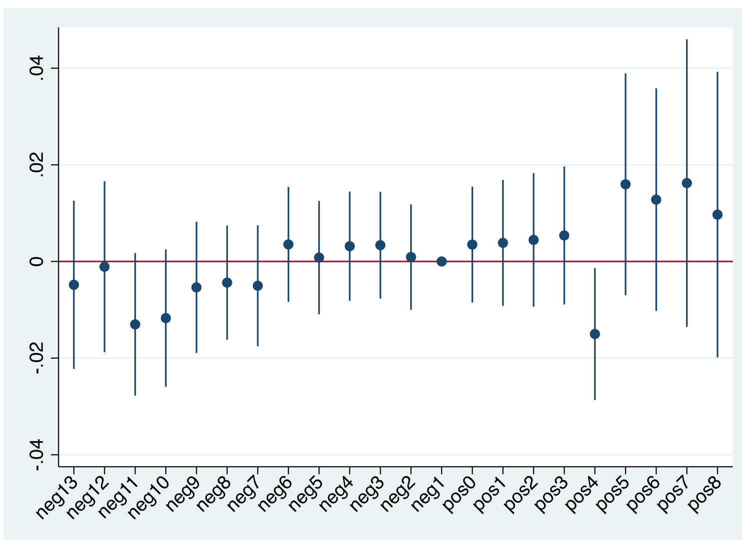
The coefficients of interest, ω_js and θs, measure the treatment effects in the years leading up to recreational marijuana legalization and the years after. The dummy for the year before RML is omitted, which normalizes the estimates of ω_j and θ to zero in that event year.The ωs are falsification tests that capture the pre-trend. Their pattern and statistical significance are a direct test of the common trends assumption. Theθs are intent-to-treat (ITT) effects of RML on the outcomes, which allow heterogeneity in the treatment effects of different years after the RML.

5 Estimates of RML's ITT Effect

Based on the previous literature, consumption of marijuana among college students has increased significantly after legalization (Miller et al., 2017; Jones et al., 2018; Koval et al., 2019; Barker and Moreno, 2021), and marijuana use is associated with increased cognitive, mental health, and substance use disorders (Ammerman et al., 2015). In this section, I study the effects of RML on two most important education outcomes of college students, retention rates and graduation rates. Estimates obtained in this section are intent-to-treat(ITT) effects of the recreational marijuana legalization.Sample used in the regression are $5,095$ U.S. postsecondary institutions from the 2008 to 2020 IPEDS, which covering both 4-year institutions and two-year and less-than-2-year institutions, as well as public institutions, private not-for-profit institutions, and private-for-profit institutions.

5.1 Retention Rates

The first set analysis is based on the retention rates. Figure 1 presents the estimates of equation (1) using college retention rates. The small estimates in the years before RML strongly support the research design by ruling out differential trends. According to the estimates of the years after RML, except for the fourth year after the legalization, estimates of the effect on retention rate are positive and statistical insignificant. Only in the fourth year after RML, college retention rate. dropped significantly by 1.5 percentage point. According to table 3, as ten-year average of retention rate mean for the treated group is 73%, state recreational marijuana legalization decreased the retention rates of institutions in the legalized states by around 2 % on average.



The graph has omitted the coefficient of "neg14" as only institutions in Rhode Island contribute to it and has a big standard error. The coefficient of neg14 and its stand error are shown in Table 1.A

Figure 1. Estimates of RML on Retention Rates

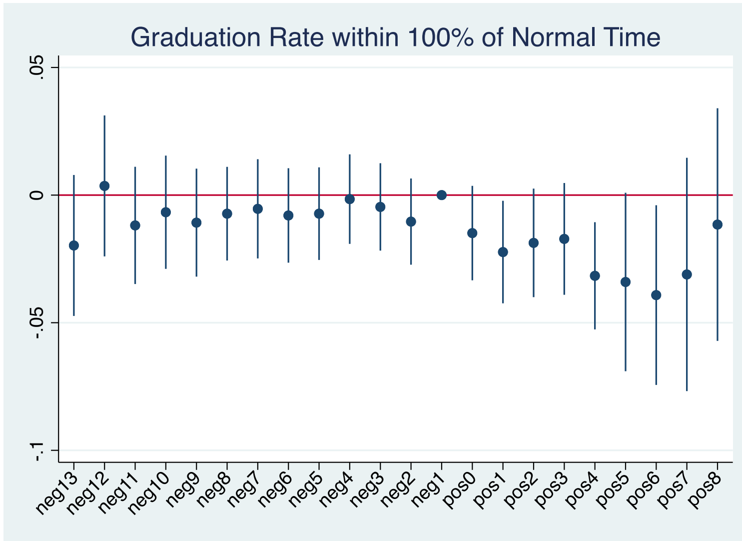
To compare the effects of RML on institutions of different levels, separate regressions are conducted by the level of institutions, and results are shown in Appendix Table 1.A and Figure 1.A. Estimates for 4-year institutions are similar to Figure 1, and effects for 2-year and less than 2-year institutions are insignificant. Thus, significant drop of the retention rates in the fourth year after RML is mainly driven by 4-year institutions in my sample.

5.2 Graduation Rates

Another education outcome studied in this paper is graduate rate. As the previous literature has shown that RML significantly increased college marijuana consumption, it is reasonable to infer a negative effect on college students' in-school behaviors, which may lower the graduation rates.

Firstly, I study the effect on the normal time graduation rates of postsecondary institutions across different levels. Figure 2 presents the regression results of equation (1). The lack of differential trends in normal-time graduation rates prior to RML supports interpreting the post-RML estimates as causal effects of the RML on graduation rates rather than pre-existing differences between treatment and control institutions. The estimates fall immediately after the state legalized recreational marijuana use, though not significantly different from zero for some years, and have an increasing trend in terms of the magnitude in the first 6 years after RML. Detailed regression results are showed in the Appendix Table 2,A. The recreational marijuana use legalization decreased the normal-time graduation rates significantly in the first, fourth and sixth year after the state legalized recreational marijuana use by 2.2 percentage points, 3.2 percentage points and 3.9 percentage points respectively. As the 10-year average mean of the normal-time graduation rate of institutions in the treat group is only 39% (Table 2), the negative effects account for around 6%, 8% and 10% respectively, which means the RML decreased the graduation rates greatly.

To further explore the negative impacts of the RML on the normal-time graduation rates, similar to the retention rates, I conducted separated regressions by the level of the institutions. Results are shown in the Appendix Figure 2.A and the regression result table is shown in the Appendix Table 2.B. The pattern of 2-year and less than 2-year institutions is similar to Figure 2, however, there aren't any significant effects on 4-year institutions. Thus, the negative effects of the RML on normal-time graduation rates are mainly driven by 2-year and less than 2-year institutions.



The graph has omitted the coefficient of "neg14" as only institutions in Rhode Island contribute to it and has a big standard error. The coefficient of neg14 and its stand error are shown in Column (1) of Table 2.A

Figure 2. Estimates of RML on Normal-time Graduation Rates

Beside the normal-time graduation rates, effects on graduation rates within 150% normal time and graduation rates within 200% of normal time present in Figure 3. Detailed regression results are showed in Appendix Table 2.A column (2) and (3). Different from normal-time graduation rates, regressions from my sample show the RML doesn't have significant effects on 150%-normal-time and 200%-normal-time graduation rates.

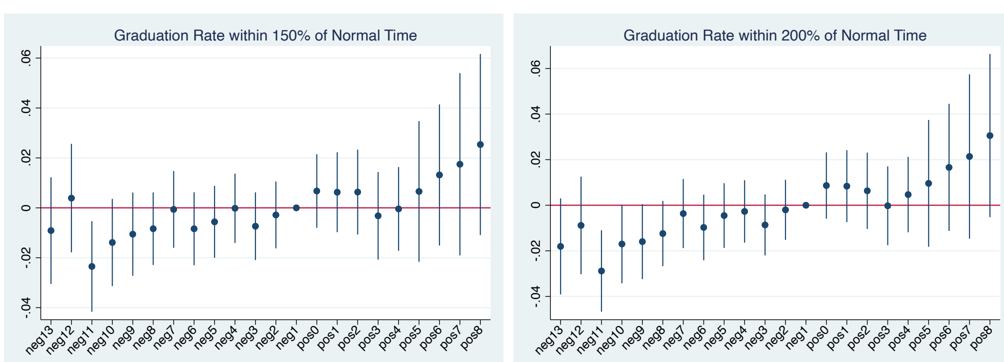


Figure 3. Estimates of RML on 150%-normal-time Graduation Rates and 200%-normal-time Graduation Rates

6 Discussions

The negative effects on retention and graduation rates are consistent with our intuitions given the positive effect of RML on marijuana consumption among college students studied in the previous literature and the negative physical and mental impacts on marijuana users. In terms of the graduation rates, the increase of the magnitude of the effect in the first several years can be explained by two reasons. First, the graduation rate of current college students need to be observed later. Even if the students began to use marijuana or increased marijuana consumption immediately after it became legalized, the effect on the graduation rate could not be observed until the their graduation year. Second, not all students get access and react to the law change immediately.. As time passing by, the spread of the information and influence from their peers make the impact of RML greater.

Comparing the impacts of RML on graduation rates by length, we saw that while the legalization of recreational marijuana use significantly decreased the proportion of students graduating within normal time, it doesn't affect the proportion of students graduating within 150%-normal-time and 200%-normal time. Thus, the RML potentially extended the college students' graduation time. Further research can be conducted on the effects of RML on the graduation time if data is available.

However, this paper has some unsolved questions.. First, it doesn't provide an explanation of why the effect on retention rate is negative and statistically different from zero only in the fourth year after RML. Second, the empirical evidence in the paper shows that while the decrease of the retention rate is manly driven by 4-year institutions, the drop of normal-time graduation rate is only caused by 2-year and less than 2-year institutions. If the RML affects students in-school behaviors in four-year institutions, we should also observe impact on the graduation rates. One possible reason is the small sample size and large standard errors. Further research can be conducted with individual-level dataset.

7 Conclusion

This paper studies the effects of recreational marijuana legalization on college education outcomes, retention rates and graduation rates using institution-level IPEDS data. Sample includes $5,095$ U.S. postsecondary institutions from the 2008 to 2020 IPEDS who have reported the retention rates and graduation rates during the period. It covers both 4-year institutions and two-year and less-than-2-year institutions, as well as public institutions, private not-for-profit institutions, and private-for-profit institutions.

Using an event study model with calender year fixed effects, state fixed effects , event-year dummies and controlling a bunch of institution characteristics, results of the study show the college retention rates dropped significantly by 1.5 percentage points in the fourth year after state legalizing the recreation use of marijuana, and the effects is mainly driven by 4-year institutions. In addition, separate regressions are conducted based on the length of the graduation time. While no significant effects are found on graduation rates within 150% normal time and 200% normal time, RML decreased the normal-time graduation rates significantly in the first, fourth and sixth year after the state legalized recreational marijuana use by 2.2 percentage points, 3.2 percentage points and 3.9 percentage points respectively, which is mainly caused by 2-year and less than 2-year institutions. Though effects are not statistically significant fro each year in the post-period, the magnitude of the effects shows a increasing trend in the first six years after legalization. Thus, I concluded that RML potentially extended college students' graduation time.

However, the study can't provide an explanation on why RML affects retention rates of 4-year institutions while affecting the normal-time graduation rates of 2-year and less than 2-year institutions. Further study can be conducted with individual-level data for more precise estimates.

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Appendix

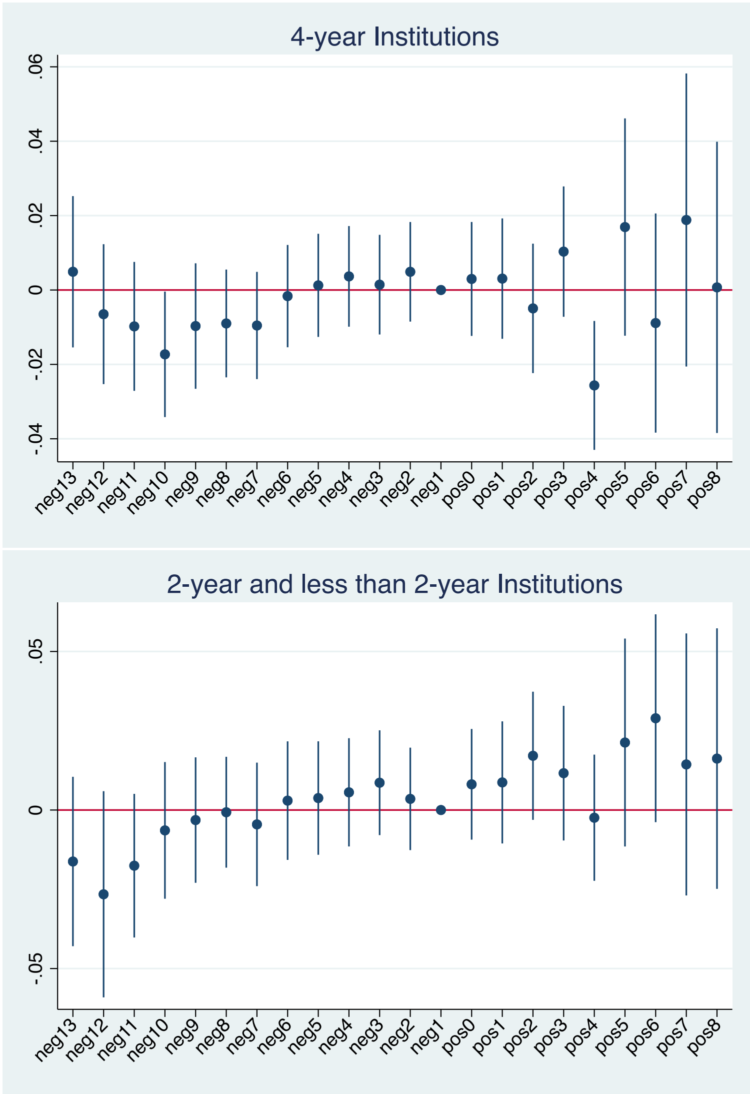

Figure 1.A Estimates of RML on Retention Rates by Level

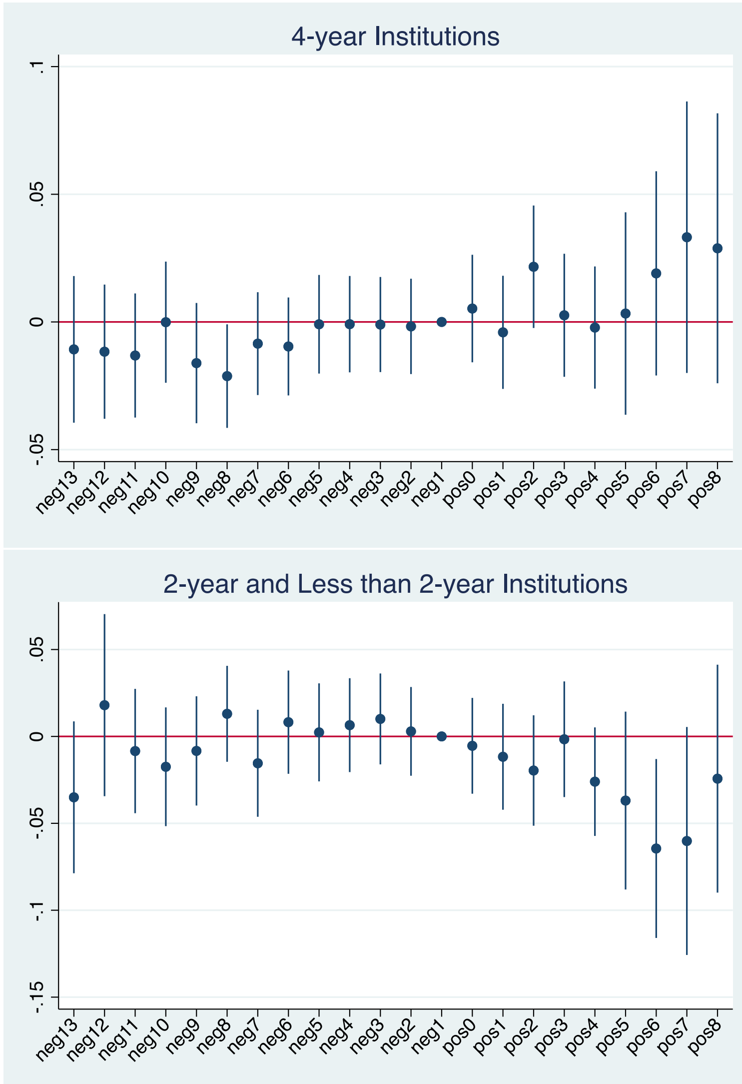

Figure 2.A Estimates of RML on Normal-time Graduation Rates by Level

Table 1.A. Estimates of RML on Retention Rates

	All Institutions	4-year Institutions	2-year and Less than 2-year Institutions
Historically Black College or University	-0.061***	-0.063***	-0.074***
	(-13.36)	(-15.11)	(-4.75)
Public	0.000	0.000	0.000
	(.)	(.)	(.)
Private Not-for-profit	0.037***	0.006**	0.055***
	(18.08)	(3.00)	(11.63)
Private For-profit	0.038***	-0.141***	0.065***
	(17.98)	(-33.92)	(24.37)
Student-to-faculty Ratio	-0.002***	-0.004***	-0.001***
	(-19.89)	(-23.67)	(-9.41)
Percent of Full-time First-time Undergraduates Receiving any Financial Aid	-0.001***	-0.002***	-0.001***
	(-17.96)	(-23.20)	(-9.55)
12-month Unduplicated Headcount	0.000***	0.000***	0.000
	(8.33)	(16.75)	(1.67)
neg14	-0.003	-0.001	-0.005
	(-0.07)	(-0.02)	(-0.07)
neg13	-0.005	0.005	-0.016
	(-0.54)	(0.47)	(-1.19)
neg12	-0.001	-0.006	-0.027
	(-0.12)	(-0.68)	(-1.60)
neg11	-0.013	-0.010	-0.018
	(-1.73)	(-1.11)	(-1.52)
neg10	-0.012	-0.017*	-0.006
	(-1.62)	(-2.01)	(-0.58)
neg9	-0.005	-0.010	-0.003
	(-0.77)	(-1.13)	(-0.31)
neg8	-0.004	-0.009	-0.001
	(-0.73)	(-1.22)	(-0.08)
neg7	-0.005	-0.010	-0.005
	(-0.79)	(-1.30)	(-0.46)
neg6	0.004	-0.002	0.003
	(0.58)	(-0.23)	(0.31)
neg5	0.001	0.001	0.004
	(0.13)	(0.18)	(0.41)
neg4	0.003	0.004	0.006
	(0.55)	(0.53)	(0.64)
neg3	0.003	0.001	0.009
	(0.60)	(0.21)	(1.02)
neg2	0.001	0.005	0.004
	(0.16)	(0.72)	(0.43)
neg1	0.000	0.000	0.000
	(.)	(.)	(.)
pos0	0.003	0.003	0.008
	(0.57)	(0.38)	(0.91)
pos1	0.004	0.003	0.009
	(0.58)	(0.37)	(0.89)
pos2	0.004	-0.005	0.017
	(0.63)	(-0.56)	(1.66)
pos3	0.005	0.010	0.012
	(0.74)	(1.15)	(1.07)
pos4	-0.015*	-0.026**	-0.002
	(-2.15)	(-2.90)	(-0.24)
pos5	0.016	0.017	0.021
	(1.36)	(1.14)	(1.27)
pos6	0.013	-0.009	0.029
	(1.09)	(-0.59)	(1.73)
pos7	0.016	0.019	0.014
	(1.07)	(0.94)	(0.68)
pos8	0.010	0.001	0.016
	(0.64)	(0.04)	(0.77)
cons	0.784***	0.926***	0.734***
	(105.49)	(94.51)	(65.06)

* p¡0.05, ** p¡0.01, *** p¡0.001

Table 2.A. Estimates of RML on Graduation Rates

	(1) 100% Normal Time GR	(2) 150% Normal Time GR	(3) 200% Normal Time GR
Historically Black College or University	-0.149***	-0.116***	-0.110***
	(-21.32)	(-20.72)	(-19.82)
Public	0.000	0.000	0.000
	(.)	(.)	(.)
Private for-profit	0.153***	0.109***	0.096***
	(48.63)	(42.74)	(38.20)
Private not-for-profit	0.075***	0.146***	0.129***
	(23.50)	(56.82)	(50.90)
Student-to-faculty Ratio	-0.007***	-0.005***	-0.005***
	(-36.70)	(-34.02)	(-34.53)
Percent of Full-time First-time Undergraduates Receiving any Financial Aid	-0.001***	-0.001***	-0.001***
	(-12.64)	(-17.32)	(-18.89)
12-month Unduplicated Headcount	-0.000***	-0.000***	-0.000***
	(-10.95)	(-7.98)	(-5.22)
neg14	0.004	-0.038	-0.070
	(0.07)	(-0.75)	(-1.42)
neg13	-0.020	-0.009	-0.018
	(-1.40)	(-0.84)	(-1.68)
neg12	0.004	0.004	-0.009
	(0.26)	(0.35)	(-0.81)
neg11	-0.012	-0.023*	-0.029**
	(-1.01)	(-2.54)	(-3.16)
neg10	-0.007	-0.014	-0.017
	(-0.59)	(-1.56)	(-1.94)
neg9	-0.011	-0.011	-0.016
	(-1.00)	(-1.24)	(-1.90)
neg8	-0.007	-0.008	-0.012
	(-0.78)	(-1.13)	(-1.70)
neg7	-0.005	-0.001	-0.004
	(-0.54)	(-0.08)	(-0.47)
neg6	-0.008	-0.008	-0.010
	(-0.84)	(-1.12)	(-1.32)
neg5	-0.007	-0.006	-0.005
	(-0.79)	(-0.76)	(-0.62)
neg4	-0.002	-0.000	-0.003
	(-0.18)	(-0.03)	(-0.39)
neg3	-0.005	-0.007	-0.009
	(-0.53)	(-1.06)	(-1.27)
neg2	-0.010	-0.003	-0.002
	(-1.21)	(-0.42)	(-0.30)
neg1	0.000	0.000	0.000
	(.)	(.)	(.)
pos0	-0.015	0.007	0.009
	(-1.58)	(0.90)	(1.17)
pos1	-0.022*	0.006	0.008
	(-2.18)	(0.77)	(1.04)
pos2	-0.019	0.006	0.006
	(-1.73)	(0.73)	(0.74)
pos3	-0.017	-0.003	-0.000
	(-1.54)	(-0.36)	(-0.03)
pos4	-0.032**	-0.000	0.005
	(-2.95)	(-0.05)	(0.55)
pos5	-0.034	0.007	0.010
	(-1.91)	(0.46)	(0.68)
pos6	-0.039*	0.013	0.017
	(-2.18)	(0.91)	(1.17)
pos7	-0.031	0.017	0.021
	(-1.33)	(0.94)	(1.16)
pos8	-0.012	0.025	0.031
	(-0.50)	(1.37)	(1.67)
cons	0.527***	0.665***	0.740***
	(45.73)	(72.83)	(82.19)

* p¡0.05, ** p¡0.01, *** p¡0.001

Table 2.B. Estimates of RML on Normal-time Graduation Rates by Level

	4-year Institutions	2-year and Less than 2-year Institutions
Historically Black College or University	-0.125***	-0.065**
	(-21.92)	(-2.71)
Public	0.000	0.000
	(.)	(.)
Private Not-for-profit	0.165***	0.107***
	(55.69)	(14.43)
Private For-profit	0.075***	0.026***
	(12.78)	(6.30)
Student-to-faculty Ratio	-0.010***	-0.004***
	(-35.55)	(-18.01)
Percent of Full-time First-time Undergraduates Receiving any Financial Aid	-0.004***	0.000
	(-38.08)	(0.46)
12-month Unduplicated Headcount	0.000***	-0.000***
	(13.63)	(-26.17)
neg14	-0.025	0.070
	(-0.43)	(0.62)
neg13	-0.011	-0.035
	(-0.73)	(-1.57)
neg12	-0.012	0.018
	(-0.87)	(0.67)
neg11	-0.013	-0.008
	(-1.06)	(-0.46)
neg10	-0.000	-0.017
	(-0.01)	(-1.00)
neg9	-0.016	-0.008
	(-1.34)	(-0.52)
neg8	-0.021*	0.013
	(-2.05)	(0.92)
neg7	-0.008	-0.015
	(-0.83)	(-0.98)
neg6	-0.010	0.008
	(-0.98)	(0.54)
neg5	-0.001	0.002
	(-0.09)	(0.16)
neg4	-0.001	0.006
	(-0.09)	(0.47)
neg3	-0.001	0.010
	(-0.11)	(0.75)
neg2	-0.002	0.003
	(-0.18)	(0.22)
neg1	0.000	0.000
	(.)	(.)
pos0	0.005	-0.005
	(0.49)	(-0.38)
pos1	-0.004	-0.012
	(-0.36)	(-0.75)
pos2	0.022	-0.020
	(1.77)	(-1.21)
pos3	0.003	-0.002
	(0.21)	(-0.10)
pos4	-0.002	-0.026
	(-0.18)	(-1.63)
pos5	0.003	-0.037
	(0.16)	(-1.41)
pos6	0.019	-0.064*
	(0.93)	(-2.46)
pos7	0.033	-0.060
	(1.22)	(-1.80)
pos8	0.029	-0.024
	(1.07)	(-0.73)
cons	0.670***	0.474***
	(48.81)	(26.45)

* p¡0.05, ** p¡0.01, *** p¡0.001